Abdul Rehman

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RESEARCH INTEREST

- Metal Recovery & Recycling
- Materials Characterization
- Sustainable Manufacturing
- Advanced Materials & Energy Material
- Welding and Joining Processes

EXPERIENCES

- **Quality Control Engineer** **Jan, 2025 – Present**

Applied Engineering *Lahore, Pakistan*

Ensured quality compliance concerns and coordinated with manufactures to enforce adherence to SNGPL, API and ASTM specifications for mild steel service line materials, polyethylene and Al. materials used in gas distribution systems.

- **R&D Lab Assistant** **Aug, 2023 – Feb 2024**

Pakistan Council of Scientific and Industrial Research *Lahore, Pakistan*

Performed ore analysis and lab scale extraction in compliance with ASTM and ISO standards, gained practical experience in hydrometallurgy and assisted in mentoring internees at mineral processing research center.

- **Associate Intern** **July 2021 – Aug 2021**

Ravi Green Engineering Pvt. Limited *Lahore, Pakistan*

Developed expertise in welding quality assurance by applying ASME codes & NDT methods to evaluate materials, fabrication processes, and WPS compliance, alongside interpretation engineering design for pressure system.

EDUCATION

- B.Sc. Metallurgy & Materials Engineering **Nov 2018 – Dec 2022**

University of the Punjab *Lahore, Pakistan*

PROJECTS

- Development and Characterization of Glass Fiber Reinforced Concrete
- Towards Sustainable Zinc Recycling: Cutting edge approaches for selective recovery from waste streams
- Cast 4 tensile specimens of LM-13 according to ASTM E8 to analyze thermal behavior, mechanical post solidification properties and casting defects
- Study of Al Bronze coating deposited by electric arc method on SS 304 for marine application
- Simulation based comparative structural analysis and failure prediction of steel vs composite leaf spring under conditional compressive Load

EXTRACURRICULAR ACTIVITIES

- **Conference:** 3rd International Conference on Trends and Research in Chemistry – 2024
- **Workshop:** 3-Days Materials Characterization Workshops – 2024
- **Workshop:** Production and Operation Management Workshop – 2024
- **Research Symposium:** All Punjab Universities Innovation - 2022
- **Research Symposium:** 1st Engineering Capstone – 2022
- **Student Ambassador:** IJT Pakistan

TECHNICAL SKILLS

- **Labs/Characterization Techniques:** Optical metallography, XRD, SEM, TGA, DSC, Mechanical testing, Extractive metallurgy
- **Software Competencies:** Fusion 360, Ansys, OriginPro, Python, AutoCAD
- **Interpersonal skills:** Research Article writing, Academic presentation, Team collaboration, Lab Protocol Management

AWARDS AND HONORS

- 1st Position in Final Year Project Presentation